

TAN Can

Applied Economics | AI for Finance & Corporate Credit Risk

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Portfolio & Web: https://7ancc.github.io/can-tan-research/

RESEARCH INTERESTS

- **AI for Finance** ; Corporate Credit Risk ; Financial NLP and Time-Series Forecasting ; Causal Machine Learning ; Applied Econometrics ; Environmental & Sustainable Finance

EDUCATION

Shandong University | School of Business Sep 2025-Jun 2028

- *Master's student in Applied Economics, Finance/Insurance* | GPA: 90.91/100 | **Rank: 8/46**
- **Major Courses:** Intermediate and Advanced Econometrics(99); Intermediate and Advanced Macroeconomics(95); Artificial Intelligence in Business(93); Microeconomic Finance Theory(90.5)
- **Selected Honors: Second-Class University Scholarship; Outstanding Individual in Social Service**

Guangdong Ocean University | School of Management Sep 2021-Jun 2025

- *Bachelor's degree in Financial Management* | GPA: 4.13/5.00 | **Rank: 1/151**
- **Major Courses:** Mathematics for Economics(97); Marketing(96); Advanced Financial Management(94); Probability and Mathematical Statistics(92); Corporate Strategy and Risk Management(93); Investments(93); Linear Algebra(90)
- **Selected Honors: Ranked 1st in the School** for postgraduate recommendation and admitted to **Shandong University** through the exam-exempt pathway; **National Scholarship**; Top 10 Outstanding Student of the University; First-Class University Scholarship (awarded annually); Outstanding Student Leader (University Level)

RESEARCH OUTPUTS

Publications

- ***Tan, C**.*, Zhong, J.*, Yang, D.*, & Huang, W. (2025). Forecasting and analysing global average temperature trends based on LSTM and ARIMA models. PLOS One, 20(9), e0330645. https://doi.org/10.1371/journal.pone.0330645 * Equal contribution; first-listed author.(JCR Q1)*
- *Li, H., He, X., **Tan, C**., Zhang, Y., Yang, D., & Li, Y. (2024). Revolutionizing healthcare: Unveiling China's healthcare efficiency quest and path to progress. Finance Research Letters, 67, 105688. https://doi.org/10.1016/j.frl.2024.105688 (SSCI | JCR Q1)*

Manuscripts Under Review | First Author

- *Environmental Fee-to-Tax Reform and Corporate Debt Default Risk: The Dual Perspectives of Internal Financial Restructuring and External Market Clearing.* | Submitted to *Securities Market Herald(Composite IF:14.775 | CSSCI)*
- *The Dark Side of Carbon Markets? ETS-Induced Reduction in Corporate Patient Capital in China's High-Pollution Industries.* | Submitted to *Ecological Economics(JCR Q1)*

TECHNICAL SKILLS & LANGUAGES

- **Programming & Data:** Python ; Stata ; SPSS ; Excel
- **AI & Machine Learning:** scikit-learn ; XGBoost ; PyTorch ; Transformers ; NLP ; LSTM ; Predictive Modeling
- **Econometrics & Quantitative Methods:** DID/DDD ; Panel Data ; DEA ; ARIMA ; Time-Series Forecasting
- **Research Tools:** CiteSpace ; Origin ; Vensim ; LaTeX ; Git/GitHub ; LLM/Agentic Workflows
- **Languages:** Mandarin Chinese (Native) · English (CET-6)

RESEARCH EXPERIENCE

***Corporate Credit Risk, Environmental Finance & Causal Machine Learning** | Independent Research* Sep 2024-Present

- Investigated the impact of China's environmental fee-to-tax reform on corporate credit risk, combining a **KMV-based Expected Default Frequency(EDF)** measure with firm panel data and **DID/DDD** designs in Stata
- Conducted fixed-effects estimation, parallel-trend/placebo tests, robustness/endogeneity checks, PSM-DID, and heterogeneity/mechanism analyses. Awarded **Outstanding Undergraduate Thesis** (96/100, 1st in the School)
- Extended the thesis using a DDD framework to examine financial restructuring and market clearing, and applied **Double/Debiased Machine Learning (DDML)** to corporate default-risk analysis

***Quantitative Modeling & Predictive Analytics** | Research Writing & Analysis Lead* Aug 2023

- **Climate Forecasting — LSTM & ARIMA:** Led data analysis and manuscript development for global-temperature forecasting using LSTM, ARIMA, cross-validation, and correlation analysis. Published in *PLOS One* as first-listed co-first author and awarded **First Prize** in the 2022 Asia-Pacific Mathematical Contest in Modeling

- **NLP & Consumer Review Analytics:** Led manuscript development and analysis for a **BERT-BiLSTM-CNN** study, including text preprocessing, review classification, sentiment prediction, keyword extraction, and human–machine review detection, achieving approximately **92%** Macro-Precision/Recall/F1
- **Risk Analytics — ARIMA & XGBoost:** Led manuscript development and empirical analysis for an aviation-risk study using ARIMA, XGBoost, statistical testing, anomaly detection, and optimization on QAR data to develop risk-warning models, earning **Second Prize** in the 2023 MathorCup Mathematical Modeling Challenge
- **Independently reproduced** LSTM/ARIMA, transformer-based NLP, and XGBoost workflows in Python

AI-Enabled Intelligent Systems & Technology Commercialization | *Deputy Project Lead* Mar 2024-Aug 2025

- **Citrus-Pest Monitoring & Treatment System:** Led market research, business-model design, financial modeling, SKU strategy, and commercialization; collaborated with the technical team on AI-enabled pest detection and early-warning applications. The project received **six awards**, including **First Prize** in the Provincial "Challenge Cup"
- **Multi-Layer Intelligent Unmanned Planting System:** Participated in an interdisciplinary smart-agriculture project integrating intelligent equipment and automated cultivation, earning **"Swan Cup" First Prize** and a **National Bronze Award** in the China International College Students' Innovation Competition
- Led the translation of **R&D outputs into product design, market strategy, and financial models**, driving **end-to-end 0-to-1 commercialization** from prototype validation to external partnerships

Consumer Analytics & Market Research | *Team Lead* Nov 2022-Apr 2023

- Led a five-member team studying recreational-fishery consumer satisfaction, combining **fieldwork and 434 valid questionnaires** with Python/Jieba/SO-PMI text analysis, SPSS, ACSI modeling, and fuzzy evaluation, translating findings into recommendations and earning **Provincial Second Prize** in the "CP Cup" Market Survey Competition

Neural Dynamics, Intelligent Detection & Intellectual Property | *Research Contributor* Oct 2023-Jul 2024

- Contributed to an interdisciplinary neural-dynamics and high-dimensional computation research stream through literature review, mathematical-method screening, data sourcing, figure preparation, research synthesis, manuscript drafting support, and patent documentation
- Supported research on the **ACGND framework** for dynamic tensor inversion, applying adaptive gradient neural dynamics to high-dimensional computational problems relevant to TensorFlow- and PyTorch-based AI workflows, with related work resulting in a **granted invention patent in hyperspectral target detection**
- Developed the ability to rapidly understand unfamiliar mathematical and computational methods, reproduce core workflows, and translate technical results into structured academic writing

Financial Modeling & Venture Commercialization | *Finance Team Member* Sep 2022-Dec 2023

- Led **financial modeling and data-driven commercial analysis**, covering accounting, financial statements, cash-flow and profitability analysis, and business planning to support pricing and commercialization; the project received **10+ national/provincial awards**, including a **National Silver Award** in the China International College Students' Innovation Competition, and was featured by CCTV-1 and 10+ mainstream media outlets

SELECTED PROFESSIONAL EXPERIENCE

Sinolink Securities Co., Ltd. | *Investment Banking, IPO Project* Jun 2026-Sep 2026

- Conducted data-intensive IPO due diligence across **4,350+** transaction records, **45** material contracts, **43,000**+ bank transactions, and **330+** shareholder dividend flows, identifying potential related-party and fund-flow risks and converting findings into **IPO disclosure and procurement-price analysis**

Zhuhai Da Heng Qin Group Co., Ltd. | *Finance Intern* Dec 2023-Mar 2024

- Prepared annual operating analyses for **two subsidiaries**, evaluating revenue, costs, cash flows, ROE, ROA, and liquidity**,** and produced management-oriented financial reports and visualizations

SELECTED INTELLECTUAL PROPERTY

- **Granted Invention Patent**: Hyperspectral Image Target Detection Method — *Co-Inventor , ZL202410627687.5*
- **Utility Model Patent:** A Financial Information Storage Device — *First Inventor, ZL202322544773.3*
- **Registered Software Copyrights:** Pest Classification & Recognition Algorithm ; Intelligent Pest Detection System

ACADEMIC LEADERSHIP

Shandong University - *President***:** Organized an **"AI for Research" seminar** and analyzed 215 survey responses